



Master Trainer Kit: Complete Student Laboratory Manual

Welcome to the Master Trainer Kit. This Boolean board contains 32 unique digital logic circuits, plus a hidden 33rd circuit (a 4-bit Arithmetic Logic Unit), all programmed into a single FPGA. You will use the on-board switches and buttons to select a circuit, provide inputs, and observe the outputs on the LEDs.

1. Hardware Layout & Controls

Before testing, familiarize yourself with the board's control zones:

- **Zone 1: Mode Selectors (SW15 to SW11)**
 - These 5 left-most switches act as a binary selector (from 00000 to 11111). Set these to choose which of the 32 normal circuits is active.
 - LED15 to LED11 will mirror your selector switches so you can visually confirm your binary mode.
- **Zone 2: The ALU Override (SW10 & LED10)**
 - Flipping SW10 UP activates the 4-bit ALU, overriding whatever is selected on SW15-11. LED10 will turn ON to warn you that the ALU is active. Keep this DOWN for normal 32-case operation.
- **Zone 3: Data Inputs & Outputs (SW9 to SW0)**
 - The right-most switches act as your binary inputs (A, B, C, Data, Select lines) for the active circuit.
 - LED9 to LED0 display the results. Any LED not required by the current circuit is automatically forced OFF.
- **Zone 4: Action Buttons (btn[3] to btn[0])**
 - Used exclusively for sequential circuits (memory, counters, shift registers) to control the flow of time:
 - **btn[0] (Clock):** Tap to pulse the clock and step the circuit forward.
 - **btn[1] (Reset):** Hold to clear the circuit's memory back to zero.
 - **btn[2] (Set):** Used only in the SR Latch to force the output to 1.
 - **btn[3] (Load):** Hold down while tapping the Clock (btn[0]) to load parallel data into the PISO shift register.

2. Basic Logic Gates (Combinational)

These circuits operate instantly. As soon as you move an input switch, the output LED will update.

Testing Procedure: Set the Mode using SW15-11. Toggle the Input switches (SW0, SW1, SW2) and compare LED0 against standard digital logic truth tables.

Mode (Binary)	Circuit Name	Input A	Input B	Input C	Output

00000	NOT Gate	SW0	-	-	LEDO
00001	2-Input AND	SW0	SW1	-	LEDO
00010	2-Input OR	SW0	SW1	-	LEDO
00011	2-Input NAND	SW0	SW1	-	LEDO
00100	2-Input NOR	SW0	SW1	-	LEDO
00101	2-Input XOR	SW0	SW1	-	LEDO
00110	2-Input XNOR	SW0	SW1	-	LEDO
00111	3-Input AND	SW0	SW1	SW2	LEDO
01000	3-Input OR	SW0	SW1	SW2	LEDO
01001	3-Input NAND	SW0	SW1	SW2	LEDO
01010	3-Input NOR	SW0	SW1	SW2	LEDO
01011	3-Input XOR	SW0	SW1	SW2	LEDO
01100	3-Input XNOR	SW0	SW1	SW2	LEDO

3. Arithmetic Circuits (Combinational)

These circuits perform 1-bit binary addition and subtraction.

Testing Procedure: Set the Mode. Toggle your A and B bits. Observe how the Sum/Difference and Carry/Borrow bits behave.

Mode (Binary)	Circuit Name	Input A	Input B	Carry/Borrow In	Output 1	Output 2
01101	Half Adder	SW0	SW1	-	LED0 (Sum)	LED1 (Carry)
01110	Full Adder	SW0	SW1	SW2 (Cin)	LED0 (Sum)	LED1 (Carry)
01111	Half Subtractor	SW0	SW1	-	LED0 (Diff)	LED1 (Borrow)
10000	Full Subtractor	SW0	SW1	SW2 (Bin)	LED0 (Diff)	LED1 (Borrow)

4. Multiplexers & Demultiplexers (Combinational)

These circuits act as digital traffic controllers, routing data based on "Select" lines.

Testing Procedure: Set your Data inputs to a distinct pattern. Toggle the Select lines and watch the data shift between outputs (DEMUX) or watch which input channel takes control of the output (MUX).

Mode (Binary)	Circuit	Data Inputs	Select Lines	Outputs
10001	2:1 MUX	SW0 (D0), SW1 (D1)	SW2 (S0)	LED0
10010	4:1 MUX	SW0 to SW3 (D0-D3)	SW4 (S0), SW5 (S1)	LED0
10011	1:2 DEMUX	SW0 (Data)	SW1 (S0)	LED0, LED1
10100	1:4 DEMUX	SW0 (Data)	SW1 (S0), SW2 (S1)	LED0 to LED3

5. Latches & Flip-Flops (Sequential)

Sequential circuits have memory. They only update when triggered by the Clock (btn[0]), except for the SR Latch which is level-sensitive.

Testing Procedure:

1. Set the Mode and press btn[1] (Reset) to clear memory.
2. Set your input switches. Notice the LEDs do not change yet.
3. Tap btn[0] (Clock) to capture the inputs and update the LEDs.

Mode (Binary)	Circuit Name	Switch Inputs	Button Controls	Outputs
10101	SR Latch	-	btn[2] (Set), btn[1] (Rst)	LED0 (Q), LED1 (Qbar)
10110	D Flip-Flop	SW0 (Data)	btn[0] (Clk), btn[1] (Rst)	LED0 (Q)
10111	T Flip-Flop	SW0 (Toggle)	btn[0] (Clk), btn[1] (Rst)	LED0 (Q)
11010	JK Flip-Flop	SW0 (J), SW1 (K)	btn[0] (Clk), btn[1] (Rst)	LED0 (Q), LED1 (Qbar)

6. Counters & Shift Registers (Sequential)

These circuits process data across 4 parallel bits.

- **Counters:** Set the mode, press btn[1] to reset, then repeatedly tap btn[0] (Clock) to watch the binary numbers count up or down.
- **Shift Registers:** Press btn[1] to reset. Input data on the switches, then tap btn[0] to push the data through the internal memory array.

Mode (Binary)	Circuit Name	Data Inputs	Button Controls	Outputs
11000	4-Bit Up Counter	-	btn[0] (Clk), btn[1] (Rst)	LED3-0
11001	4-Bit Down	-	btn[0] (Clk),	LED3-0

	Cnt		btn[1] (Rst)	
11011	SISO Shift Reg.	SW0 (Serial In)	btn[0] (Clk), btn[1] (Rst)	LED0
11100	SIPO Shift Reg.	SW0 (Serial In)	btn[0] (Clk), btn[1] (Rst)	LED3-0
11101	PISO Shift Reg.	SW3 to SW0 (Parallel In)	Hold btn[3] + tap btn[0] to load	LED0
11110	PIPO Shift Reg.	SW3 to SW0 (Parallel In)	btn[0] (Clk), btn[1] (Rst)	LED3-0
11111	Ring / Johnson	SW0 (0=Ring, 1=Johnson)	btn[0] (Clk), btn[1] (Rst)	LED3-0

(Note on Case 31: If you switch modes between Ring and Johnson using SW0, you must press btn[1] to reset the starting pattern before tapping the clock).

7. Special Feature: 4-Bit ALU (Override Mode)

The board features a hidden 33rd circuit: a complete 4-bit Arithmetic Logic Unit (ALU).

How to Activate:

Flip **SW10** UP. LED10 will turn ON. The 5 left-most mode selector switches will be temporarily ignored, and the board is now an ALU.

ALU Input Mapping:

- **SW3 to SW0:** Input A (4-bit binary number)
- **SW7 to SW4:** Input B (4-bit binary number)
- **SW9 & SW8:** Operation Selector (Opcode)
 - 00: ADD (A + B)
 - 01: SUBTRACT (A - B)
 - 10: Bitwise AND
 - 11: Bitwise OR

ALU Output Mapping:

- **LED3 to LED0:** 4-Bit Result
- **LED4:** Carry-Out (for ADD) or Borrow-Out (for SUBTRACT)

To exit ALU Mode: Simply flip SW10 back DOWN. The board will instantly return to whichever of the 32 standard cases is selected on SW15-11.